

attention function

$$1.4 \quad Y = \text{attention}(X, K, V), P \text{ permutation matrix} \\ \Rightarrow PY = \text{attention}(PX, K, V)$$

2. multi-head attention

$$2.2 \quad Y = \text{multihead}(x^T, K, V)$$

$$= \text{concat}(\text{head}_1, \dots, \text{head}_h) W^0$$

$$\text{where } \text{head}_i = \text{attention}(x^T W_i^Q, K W_i^K, V W_i^V)$$

$$x, y \in \mathbb{R}^{d_{\text{model}}}, K, V \in \mathbb{R}^{n \times d_{\text{model}}}, \text{head}_i \in \mathbb{R}^{n \times d_v}$$

$$W_i^Q, W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$$

$$W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}, \quad \forall i = 1, \dots, h$$

$$W^0 \in \mathbb{R}^{h d_v \times d_{\text{model}}}$$

$$2.3 \quad Y = \text{multihead}(X, K, V)$$

$$= \text{concat}(\text{head}_1, \dots, \text{head}_h) W^0$$

$$\text{head}_i = \text{attention}(X W_i^Q, K W_i^K, V W_i^V)$$

$$X, Y \in \mathbb{R}^{m \times d_{\text{model}}}, \text{head}_i \in \mathbb{R}^{m \times d_v}$$

$$2.4 \quad Y = \text{multihead}(X, K, V), P \text{ perm matrix} \Rightarrow PY = \text{multihead}(PX, K, V)$$

1. dot-product attention

$$1.2 \quad Y = \text{attention}(x^T, K, V)$$

$$= \text{softmax}(x^T K^T) V$$

$$\text{here } x, y \text{ vector}, x \in \mathbb{R}^{d_k}, y \in \mathbb{R}^{d_v}$$

$$K = \begin{pmatrix} R^{(1)^T} \\ \vdots \\ R^{(n)^T} \end{pmatrix} \in \mathbb{R}^{n \times d_k}$$

$$V = \begin{pmatrix} v^{(1)^T} \\ \vdots \\ v^{(n)^T} \end{pmatrix} \in \mathbb{R}^{n \times d_v}$$

$$1.3 \quad Y = \text{attention}(X, K, V)$$

$$= \text{softmax}(X K^T) V$$

$$\text{here } X = \begin{pmatrix} x^{(1)^T} \\ \vdots \\ x^{(m)^T} \end{pmatrix} \in \mathbb{R}^{m \times d_k}$$

$$Y = \begin{pmatrix} y^{(1)^T} \\ \vdots \\ y^{(m)^T} \end{pmatrix} \in \mathbb{R}^{m \times d_v}$$